

# Zainab Khalil

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## EDUCATION

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### FAST NUCES

Bachelor of Computer Science | GPA: 3.74/4.0 | Dean's List (2023–2025)

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Machine Learning, Artificial Intelligence, Software Engineering

Lahore, Pakistan

Expected May 2027

### SICAS

A-Levels | 4 A\* (Mathematics, Physics, Biology, Chemistry)

3<sup>rd</sup> Position in Punjab (Best Across 3 A-Level Subjects) | Top in Pakistan for Biology (O-Levels)

Lahore, Pakistan

Aug 2021 – Jun 2023

## WORK EXPERIENCE

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### Software Engineer Intern – AICE XPERT

Jan 2026 – Apr 2026

- Expanded a live B2B SaaS platform to 5K+ professionals by shipping full-stack features across a Next.js/Express/Supabase stack: card management, QR customisation, and NFC integration.
- Built a live phone-preview component with instant QR rendering (20+ dot shapes, gradient fills, custom logos), eliminating all third-party QR tool dependencies.
- Shipped a cross-platform analytics pipeline tracking card taps, geo-location, device breakdowns, and link clicks, directly informing the premium upsell strategy.
- Engineered 10+ RESTful API endpoints enabling dynamic data flow between frontend and backend within a 4-member Agile team.

### Software Engineering Intern – INFOTECH

Jul 2025 – Aug 2025

- Architected 5+ microservices using Spring Boot, applying SOLID principles and clean architecture for maintainability and scalability.
- Implemented service discovery via Netflix Eureka across distributed services, enabling dynamic routing and fault-tolerant communication.
- Centralized routing via an API Gateway, consolidating 10+ endpoints and improving system observability, security, and traceability.

## PROJECTS

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### ResuFlow – AI-Powered CV-to-Portfolio Pipeline | *React, Node.js, MongoDB, Gemini API*

- Built an end-to-end pipeline transforming unstructured CV data into deployed portfolio websites via AI-driven extraction.
- Leveraged Gemini API to convert free-form text into validated JSON schemas; automated deployment via Vercel APIs with integrated QR code generation (Level H error correction) for one-click hosting and sharing.

### Android Housing Service App | *Java, Firebase, Cloudinary*

- Developed a full-stack Android app supporting tenant and landlord workflows with authentication and real-time sync.
- Managed 100+ property listings via Firebase Realtime Database and integrated AI-based recommendations for improved property discovery.

### Mood-Based Music Recommendation System | *Node.js, SQL Server, Spotify API*

- Built a backend-driven system mapping user mood inputs to dynamically generated Spotify playlists (10+ songs/session).
- Designed a normalized relational schema spanning 10+ tables for efficient storage and retrieval of user and playlist data.

## TECHNICAL SKILLS

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|               |                                                                                      |
|---------------|--------------------------------------------------------------------------------------|
| Languages     | C++, Java, JavaScript, Assembly (x86)                                                |
| Frameworks    | React, Next.js, Node.js, Spring Boot, Android (Java)                                 |
| Databases     | SQL Server, MySQL, MongoDB, Firebase Realtime Database                               |
| Core Concepts | Data Structures & Algorithms, OOP, System Design, Microservices Architecture         |
| Tools         | Git, Postman, REST APIs, Gemini API, Spotify API, Cloudinary, Netflix Eureka, Vercel |